Recap: Cournot Model

- 2 Firms → identical goods

Demand: $P = 100 - Q$ where $Q = q_1 + q_2$

Cost: Each firm has $TC = 10q \rightarrow MC = 10$

Best Response Functions: $q_1^* = \frac{90 - q_2}{2}$, $q_2^* = \frac{90 - q_1}{2}$

Solve together: Cournot Equilibrium: $q_1^* = 30$, $q_2^* = 30$, $P = 40$

Firm profits: $\pi_1 = 900$, $\pi_2 = 900$

if these firms cooperate → cartel

operate like a monopolist $\left.\begin{array}{l} Q = 45 \\ P = 55 \end{array}\right\}$ each firm produces $q = 22.5$

total profit $= 2025 \rightarrow$ each firm gets $\pi_1 = 1012.5$
$\pi_2 = 1012.5$

what about cheating on the cartel? → choosing a best response

cheating firm produces more output and makes even more profit

(cheater) $\pi_1 = 1139$
$\pi_2 = 759$

If we imagine that firms can "cooperate" (cartel) or "compete" (best response)

we end up with 4 possible outcomes

Firm 1's profits

price fixing phases of the moon

Firm 1 \ Firm 2	compete	cooperate
compete	\$900	\$1139
cooperate	\$759	\$1012.5

this cartel would not survive

→ both firms always have incentive to compete

what if firms don't choose output simultaneously?

Stackelberg Model

what if one firm moves first?

Timing

$\boxed{\text{Firm 1 chooses } q_1} \longrightarrow \boxed{\text{Firm 2 observes } q_1 \text{ and chooses } q_2} \longrightarrow \boxed{q_1 + q_2 = Q \text{ determines price \& profits}}$

ex. Each firm has $TC = 10q$ $(MC = 10)$ and market demand is $P = 100 - Q$ where $Q = q_1 + q_2$

in Cournot, this leads to best response functions

$$q_1 = \frac{90 - q_2}{2} \qquad q_2 = \frac{90 - q_1}{2}$$

firm 1 should anticipate the response of firm 2 to their output

if firm 2 chooses a best response, Firm 1's revenue is:

$$TR = P \cdot q_1 = (100 - q_1 - q_2) \cdot q_1$$

plug in firm 2's best response directly here

$$= \left(100 - q_1 - \frac{90 - q_1}{2}\right) \cdot q_1$$

$$= \left(55 - \frac{q_1}{2}\right) \cdot q_1$$

$$= 55q_1 - \frac{q_1^2}{2}$$

Total Revenue: $55q_1 - \frac{q_1^2}{2}$, so $MR = 55 - q_1$

Firm 1 maximizes profit with $MR = MC$

$$55 - q_1 = 10 \Rightarrow q_1^* = 45$$

and firm 2 will have to respond with:

$$q_2 = \frac{90-45}{2} \Rightarrow q_2^* = 22.5$$

so in equilibrium: $P = 100 - 45 - 22.5 = 32.5$

Firm 1's profit: $\pi_1 = P \cdot q_1 - 10 \cdot q_1 = (32.5)(45) - (10)(45) = 1012.5$

Firm 2's profit: $\pi_2 = P \cdot q_2 - 10 \cdot q_2 = (32.5)(22.5) - (10)(22.5) = 506.25$

First Mover Advantage: Firm 1 expands output and captures market share

Firm 2 is forced to respond with lower output

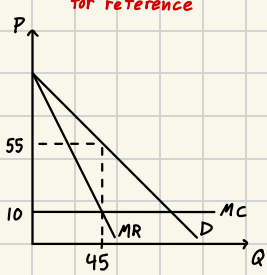
Bertrand Model

here, firms strategically set prices (identical goods)

→ consumers buy from lowest price seller
→ if prices are the same, firms split the market

ex. Demand: $P = 100 - Q$, two firms strategically set prices, each has $TC = 10q$ ($MC = 10$)

monopoly picture for reference



equilibrium: both firms set a price that is a best response to the other

<u>Best Responses</u> (Cases for firm 1)

1) if firm 2 sets $P_2 > 55$
   → firm 1's best response is to set $P_1 = 55$ (captures whole market earns monopoly profit)

2) if firm 2 sets P_2 between 55 and 10 (say $P_2 = 30$)
   → firm 1's best response is to set P_1 a tiny amount below P_2 (i.e. 29.99) (captures whole market)

3) if firm 2 sets $P_2 = 10$ (MC)
   → no incentive to undercut
   → $P_1 < 10$ (lose money)
   → $P_1 > 10$ (sell zero)
   → firm 1 will choose $P_1 = 10$ (split market, break even)

Bertrand Equilibrium: both firms set $P = MC = 10$